Shelf	Method	Why we might use it	Main caution
Discover	Direct native corpus interrogation	Baseline method; interrogate original files directly without transforming the source. Strong for traceability and reproducibility.	Search design can still miss evidence; “not found” is not “does not exist.”
Discover	Exact text / phrase search	Locate names, phrases, transaction terms, unusual wording or known indicators.	Vocabulary dependence; synonyms and spelling variants can defeat it.
Discover	Boolean / proximity search	Combine concepts, actors, dates and terms to narrow relevance.	Complex expressions may unintentionally exclude material.
Discover	Regular expressions	Find structured patterns such as IDs, values, dates, phone numbers or codes.	Poor patterns produce false positives and negatives.
Discover	Semantic search / embeddings	Find conceptually related material despite different wording.	Similarity does not equal evidential relevance.
Discover	RAG	Retrieve relevant material for AI synthesis or questioning. Useful for repeated semantic interrogation.	Retrieval introduces judgement; source and derived retrieval layer must remain distinguishable.
Discover	Full-context / direct LLM reading	Allow AI to reason directly over large collections where feasible.	Context capacity does not guarantee recall, completeness or correct interpretation.
Discover	LLM-assisted corpus exploration	Generate candidate themes, people, sequences or unusual subjects from natural-language inquiry.	Treat as lead generation until verified against primary records.
Discover	Topic modelling	Identify recurring themes without predefining vocabulary.	Mathematical topics are not findings of fact.
Discover	Clustering	Group similar documents or communications to reveal unknown patterns.	Cluster boundaries may be unstable or arbitrary.
Discover	Classification	Categorise records into defined investigative classes.	Requires validation and may encode assumptions.
Discover	Technology-Assisted Review (TAR)	Rank or categorise documents based on human-coded examples; useful at very large scale.	Training, validation and stopping criteria must be defensible.

Shelf	Method	Why we might use it	Main caution
Discover	Continuous Active Learning	Continually improve review prioritisation from reviewer decisions.	Depends heavily on coding quality.
Discover	Summarisation	Compress long document families or threads for review.	May lose qualifiers, contradictions or uncertainty.
Discover	Question answering over evidence	Ask focused evidential questions across the corpus.	Question wording can induce confirmation bias.
Reconstruct	Thread reconstruction	Rebuild email conversations from replies, forwards and quoted material.	Threads may omit telephone calls, meetings or deleted messages.
Reconstruct	Chronology / timeline reconstruction	Establish what happened and in what order. Central to knowledge, intent and causation questions.	Timestamp and time-zone issues require care.
Reconstruct	Sequence / event-chain analysis	Trace information → instruction → decision → action → consequence.	Sequence does not itself establish causation.
Reconstruct	Hanging-thread / missing-record analysis	Identify references to calls, meetings, attachments or events for which no separate record is found.	Absence must be bounded by the search performed.
Reconstruct	Named-event reconstruction	Trace a project, deal, meeting or incident across custodians and records.	Event naming may vary across participants.
Reconstruct	Knowledge-state reconstruction	Establish what evidence suggests a person knew, and when.	Receiving a communication does not prove it was read or understood.
Reconstruct	Decision reconstruction	Reconstruct what information was available when a decision was made.	Later knowledge must not be projected backwards.
Reconstruct	Process / control-path reconstruction	Establish the normal approval or operational pathway.	"Normal" must itself be evidenced.
Reconstruct	Deviation-from-process analysis	Identify missing approvals, bypassed controls or unusual routes.	Deviations may be legitimate exceptions.
Reconstruct	Flow-of-funds analysis	Trace financial benefit, movement or loss where transaction data exists.	Email may only refer to financial activity rather than record it directly.

Shelf	Method	Why we might use it	Main caution
Relate	Entity extraction	Identify people, organisations, projects, accounts and other entities.	Extraction errors may merge or split identities.
Relate	Entity resolution	Establish whether aliases, addresses or name variants refer to the same actor.	Errors contaminate downstream analysis.
Relate	Relationship / social-network analysis	Identify communication patterns, central actors, intermediaries and unusual groups.	Communication frequency does not establish culpability.
Relate	Graph analysis	Represent relationships among people, messages, transactions and events.	Graph appearance can give false impressions of significance.
Relate	Link analysis	Trace connections between actors, organisations, transactions and communications.	Association is not participation.
Relate	Organisational-role analysis	Establish authority, reporting lines and responsibilities.	Formal structures may differ from actual operating practice.
Relate	Communication-volume analysis	Identify bursts, sudden new relationships or unusual silence.	Innocent organisational events may explain patterns.
Examine	Metadata analysis	Examine timestamps, sender/recipient information, paths, IDs and document properties.	Metadata interpretation depends on the generating system.
Examine	Email header / transmission analysis	Reconstruct transmission routes, duplicates and technical history.	Headers may be incomplete or altered.
Examine	Diplomatics	Examine documentary identity, authenticity, form, provenance, function and transmission.	Specialist method; not required for every question.
Examine	Provenance / chain-of-custody analysis	Establish origin, acquisition and subsequent treatment of evidence.	Historical custody may be incomplete.
Examine	Document-family analysis	Connect emails, attachments, drafts, finals, forwards and copies.	Incorrect grouping may distort chronology or authorship.
Examine	Duplicate / near-duplicate analysis	Identify repeated or slightly altered documents and communications.	Small differences may be evidentially important.

Shelf	Method	Why we might use it	Main caution
Examine	Version comparison / document differencing	Identify exactly what changed between document versions.	Genuine version relationships must first be established.
Examine	Cross-mailbox comparison	Compare different custodians' copies of the same event or message.	Mailboxes may have differing preservation histories.
Examine	Stylometry / authorship attribution	Examine whether writing characteristics support or challenge authorship.	Probabilistic and weak on short messages.
Examine	Linguistic / discourse analysis	Examine euphemism, hedging, terminology shifts and narrative framing.	Highly contextual and prone to overinterpretation.
Examine	Code-word / euphemism discovery	Identify unusual recurring terminology and infer meaning from context.	Ordinary jargon may appear suspicious.
Test	Statistical anomaly detection	Identify unusual timing, amounts, participants, volumes or patterns.	Anomaly is a lead, not proof.
Test	Benford's Law / digit analysis	Identify unusual numerical distributions in appropriate financial datasets.	Only suitable for particular classes of data.
Test	Threshold / round-number analysis	Identify transactions around control or approval thresholds.	Requires knowledge of relevant controls.
Test	Contradiction detection	Find materially inconsistent statements or records.	Difference does not necessarily equal contradiction.
Test	Corroboration search	Seek independent support for an assertion or event.	Multiple copies of one assertion are not independent corroboration.
Test	Disconfirmatory / exculpatory search	Actively seek evidence that undermines the working theory.	Must be genuinely planned and not superficial.
Test	Competing-hypothesis analysis	Test multiple explanations against the same evidence.	Hypotheses must remain stable enough to compare.
Test	ACH — Analysis of Competing Hypotheses	Structured comparison of evidence against alternative explanations.	Can imply false precision where evidence quality is weak.

Shelf	Method	Why we might use it	Main caution
Test	Negative-space analysis	Examine expected-but-absent records, actors or communications.	Particularly vulnerable to over-inference.
Test	Temporal gap analysis	Identify unexplained periods where expected activity disappears or changes channel.	Holidays, retention rules or technical issues may explain gaps.
Test	Statistical sampling	Test prevalence, recall or supposedly irrelevant populations.	Sampling design must match the claim being tested.
Test	External-source corroboration	Compare corpus evidence with filings, public records, transaction records or other sources.	External material requires separate provenance control.
Test	Sentiment / tone-change analysis	Identify abrupt linguistic changes that may justify closer review.	Weak evidence of misconduct; primarily lead generation.
Test	Intent inference	Examine patterns potentially relevant to knowledge or intention.	Intent is generally inference, not observation.
Judge	Legal-element mapping	Map evidence to the elements of a defined offence or cause of action.	Premature use may distort exploratory investigation.
Judge	Evidential / issue matrix	Map evidence, actors, chronology, supporting material, contradictions and gaps against issues.	Framework can constrain what investigators notice.
Judge	AHP	Make explicit comparative judgements where alternatives must be ranked against criteria.	Pairwise comparisons are judgements, not primary evidence.
Judge	ANP	Analyse interconnected criteria, actors and effects where simple hierarchy is inadequate.	Considerable complexity; unnecessary for simple issues.
Judge	Bayesian / likelihood reasoning	Examine how individual evidence alters support for competing propositions.	Numerical precision may exceed what evidence permits.
Judge	Witness / interview preparation	Use documentary analysis to identify matters requiring explanation or challenge.	Documents do not substitute for testimony.
Judge	Cross-examination sequence analysis	Build evidence-backed chronological propositions for questioning.	Must distinguish investigation from advocacy.

Shelf	Method	Why we might use it	Main caution
Assure	Human linear review	Validate high-value documents and automated conclusions through direct reading.	Expensive, inconsistent and subject to fatigue at scale.
Assure	Independent replication	Have another investigator or AI repeat the same method and compare results.	Requires preserved parameters and baseline evidence.
Assure	Alternative-method replication	Investigate the same question using a different method and test convergence/divergence.	Different methods may legitimately produce different populations.
Assure	Sensitivity analysis	Vary terms, thresholds, time windows or assumptions to test whether findings persist.	Must be bounded to avoid endless testing.
Assure	Red-team / adversarial review	Deliberately attack the emerging finding and search for unsupported assumptions.	Reviewer must have genuine independence.
Assure	Reproducibility testing	Determine whether another environment can reproduce the analytical result.	Requires complete method records.
Assure	Repeatability testing	Re-run the same procedure under the same conditions and check consistency.	Repetition of a flawed method merely reproduces the flaw.

Discover → Reconstruct → Relate → Examine → Test → Judge → Assure

And importantly, they do **not** have to operate sequentially. An investigation might go:

Discover → Reconstruct → Examine → Discover again → Test → Reconstruct → Judge → Assure.

That fits Dossier Space rather nicely: **the shelves organise the available methods; CIPDA governs why and when we take one off the shelf; ROMER records what happened when we used it.**